MIL-O-37272A
22 June 1976
SUPERSEDING
MIL-O-37272
10 July 1975

MILITARY SPECIFICATION

OXYTRIPHYLLINE AND GUAIFENESIN TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Oxtriphylline and Guaifenesin Tablets (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (see 6.3)

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Oxtriphylline and Guaifenesin Tablets and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein:

S2. Classification. Shall be Type I, class 2.

The following is added to this paragraph:

Each tablet shall contain 200 mg of Oxtriphylline and 100 mg of Guaifenesin within the designated assay limits for the tablets.

The tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of Oxtriphylline and Guaifenesin when assayed as specified in 4.3.1 and 4.3.2.

The tablets shall comply with the identity test specified in 4.3.3.

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S6.4.9.1 Disintegration. Tablets shall disintegrate in not more than 25 minutes, using USP method with simulated gastric fluid T.S. maintained at 37° ± 2° as the immersion fluid.

3.2 Ingredients. All ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.3 Expiration dating period. The expiration dating period shall be 60 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

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4.3 Tests.

4.3.1 Assay for Oxtriphylline in the tablets.

NOTE: In this assay use chloroform recently saturated with purified water.

Reagents:

1 in 10 solution of Glacial Acetic Acid in Chloroform.

1 in 100 solution of Glacial Acetic Acid in Chloroform.

0.67 M Tribasic Sodium Phosphate.

Preparation of Chromatographic Column: Pack a pledget of fine glass wool in the base of a chromatographic tube (approximately 22 x 250-mm tube) with the aid of a tamping rod. To 500 mg of purified siliceous earth in a 100-ml beaker pipet 0.3 ml of 0.67 M tribasic sodium phosphate, and incorporate by kneading thoroughly to a uniform mixture. Transfer to the tube and pack lightly with the tamping rod.

Preparation of Standard Solution: Weigh accurately about 75 mg of NF Oxtriphylline Reference Standard into a 200-ml volumetric flask. Dissolve in and dilute to volume with a 1 in 100 solution of glacial acetic acid in chloroform. Pipet 3 ml of this solution to a 100-ml volumetric flask, dilute to volume with 1 in 100 solution of glacial acetic acid in chloroform, and mix.

Preparation of Sample Solution: Weigh accurately 20 tablets and determine the average tablet weight. Transfer into a mortar and reduce to a fine powder. Weigh a portion of the powder, equivalent to about 75 mg of oxtriphylline, into a 100-ml beaker, add 2.0 ml of 0.67 M tribasic sodium phosphate, and mix well. Add 3 gram of purified siliceous earth, mix thoroughly, and transfer quantitatively to the chromatographic column in three portions, packing each portion uniformly. Wash the column with three 25-ml portions of chloroform, allowing each portion to enter the column before adding the next. Collect the washings in a 200-ml volumetric flask and save for the assay of Guaifenesin. Elute the Theophylline with 5 ml of a 1 in 10 solution glacial acetic acid in chloroform, followed by 150 ml of a 1 in 100 solution of glacial acetic acid in chloroform collecting the eluates in a 200-ml volumetric flask. Dilute to volume with a 1 in 100 solution of glacial acetic acid in chloroform. Pipet 3 ml of this solution into a 100-ml volumetric flask, dilute to volume with a 1 in 100 solution of glacial acetic acid in chloroform and mix.

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Procedure: Concomitantly determine the absorbance of the sample (Au) and standard (As) solutions in a suitable spectrophotometer, at a wavelength of maximum absorbance, about 277 mu, using matched 1-cm silica cells. Use a 1 in 100 solution of glacial acetic acid in chloroform as a blank. Where S equals standard weight in mg, SW equals sample weight in mg and TW equals the average tablet weight in mg.

Calculation:

$$\text{mg Oxtriphylline per tablet} = \frac{Au}{As} \times \frac{S}{SW} \times TW$$

4.3.2 Assay for Guaifenesin in the tablets.

Preparation of Standard Solution: Weigh accurately about 40 mg of N.F. Guaifenesin Reference Standard into a 200-ml volumetric flask. Dissolve in and dilute to volume with chloroform. Pipet 25 ml of this solution into a 100-ml volumetric flask, dilute to volume with chloroform, and mix.

Preparation of Sample Solution: Dilute the chloroform solution saved in the assay for Oxtriphylline to volume (200 ml) with chloroform and mix. Pipet 25 ml of this solution into a 100-ml volumetric flask, dilute to volume with chloroform and mix.

Procedure: Concomitantly determine the absorbance of the sample (Au) and standard (As) solutions in a suitable spectrophotometer, at a wavelength of maximum absorbance, about 276 mu using matched 1-cm silica cells. Set the instrument to zero with chloroform.

Where S equals the standard weight in mg, SW equals the sample weight in mg and TW equals the average tablet weight in mg.

Calculation:

$$\text{mg Guaifenesin per tablet} = \frac{Au}{As} \times \frac{S}{SW} \times TW$$

4.3.3 Identification. The presence of Oxtriphylline and Guaifenesin in the finished tablets shall be determined as follows:

4.3.3.1 The ultraviolet absorption spectrum obtained in the assay for Oxtriphylline shall exhibit maxima and minima at the same wavelengths as that of a similar preparation of N.F. Oxtriphylline Reference Standard.

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4.3.3.2 The ultraviolet absorption spectrum obtained in the assay for Guaifenesin shall exhibit maxima and minima at the same wavelength as that of a similar preparation of N.F. Guaifenesin Reference Standard.

4.4 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

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5.2.1 Unit of issue. One bottle containing one hundred (100) tablets as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred ninety-two (192) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred ninety-two (192) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

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5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the national stock number.

NOTES: (1) The national stock number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the national stock number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings shall be shown as specified in MIL-STD-129. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking shall be shown as specified in MIL-STD-129. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this spedification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

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6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-880-4948	OXTRIPHYLLINE AND GUAIFENESIN TABLETS, 100s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1623

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-O-37272A OXTRIPHYLLINE AND GUAIFENESIN TABLETS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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